

Simple Linear Regression

STAT 301 - Lecture 01

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Welcome to STAT 301

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STAT 301 - Intro

- In most of STAT 201, we dealt with estimating and hypothesis testing one parameter.
- In this course, you will study the association **between** variables.
- We will explore the world of explanatory and predictive modelling.
- You'll learn about a variety of different models, how to interpret the results and evaluate model performance.

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Let's explore our Canvas page

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Relations between variables

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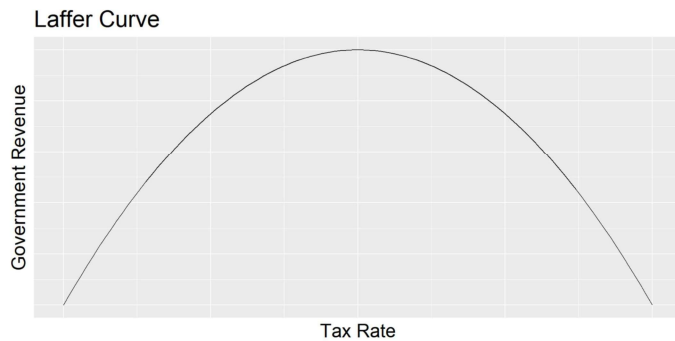
Relations

- In most of STAT 201, we focused on the estimation of one parameter of a population:
 - the average income of a group of a population;
 - the median diameter of certain trees;
- Often, we are interested in how a variable relates to other variables:
 - What do we expect to happen with the youth unemployment rate if there's an increase in minimum wage?

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Example 1: Laffer curve

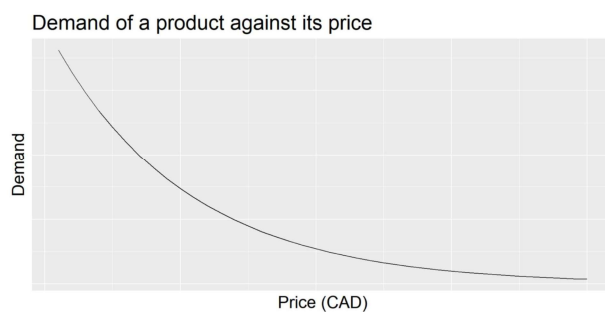
- The Laffer curve relates the tax rate and the government revenues:



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Example 2: Price vs demand

- A well-established fact in Economy is that the more expensive a product is, the smaller the demand for that product.



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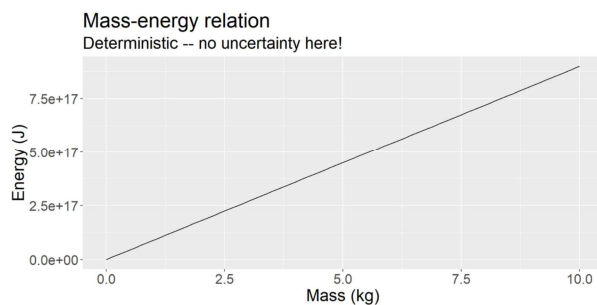
Deterministic Relations

- In a deterministic relationship, the value of one variable is entirely determined by the value of another variable.
- There is no uncertainty!

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Deterministic Relations: Example 1

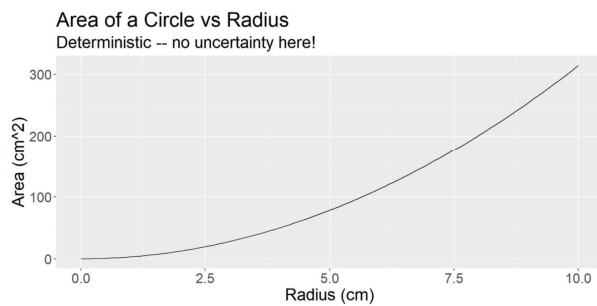
- Einstein's mass-energy relation: $E = m \times c^2$
 - c is just a constant (speed of light)



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Deterministic Relations: Example 2

- A circle's area-radius relation: $A = \pi r^2$.
 - π is just a constant.



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Stochastic Relations

- Stochastic relations refer to situations in which the outcome of a given input cannot be precisely predicted.
- There's *uncertainty*!
- The uncertainty might be due to other variables not being considered or even noise.

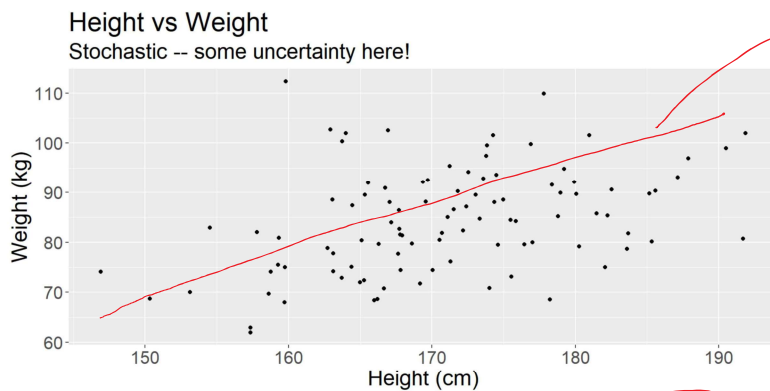
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Stochastic Relations: Example 1

- Height and weight of individuals: taller people tend to weigh more, but there is considerable variability.
- Other factors like age, gender, and lifestyle can influence weight.
- It is impossible to accurately determine a person's weight based solely on height.

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Stochastic Relations: Example 1



wrong

40%

$R^2 = 0.4$

IQ

EQ?

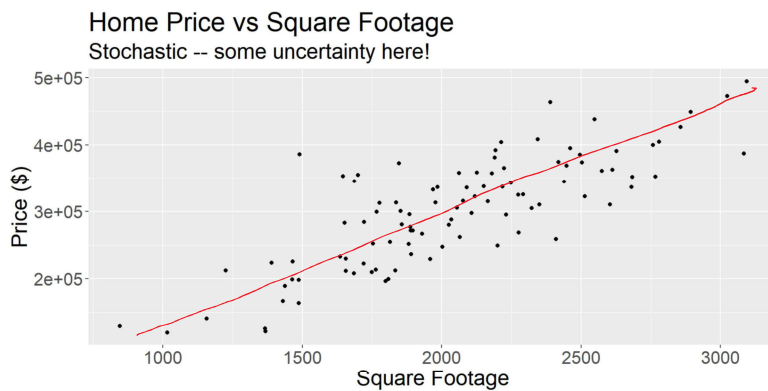
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Stochastic Relations: Example 2

- Home price and square footage: larger homes tend to cost more, but there's considerable variability.
- Other factors such as location, age of the home, and market conditions can influence the price.

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Stochastic Relations: Example 2



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Stochastic Relations: the error term

- A stochastic relation typically includes a random error term (ε) to account for the variability in the response. For instance,

$$\text{Weight} = \beta_0 + \beta_1 \times \text{Height} + \varepsilon$$

→ uncertainty

- Now, the model equation allows for two people with the same height to have different weight values;

→ Q: Why do we first consider a linear relationship?

$$y = f(x)$$
$$y = \beta_0 + \beta_1 x$$

↑
Simplest

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Shape of relation

- We often categorize a relation based on the form of the model equation:
 - Linear Relation: $\text{Weight} = \beta_0 + \beta_1 \times \text{Height} + \varepsilon$
 - Quadratic Relation: $A = \pi \times r^2$
 - Exponential Relation: $N = N_0 e^{-\lambda t} \varepsilon$
- Our focus will be exclusively on stochastic relations.

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Terminology

- We will denote our response variable by Y , also known as:

- Output variable
- Dependent variable (avoid!)

- Predictor X , also known as

- feature
- input variable
- covariate
- regressor
- independent variable (avoid!)

*variable of interest, e.g. grade
depends on study objective.*

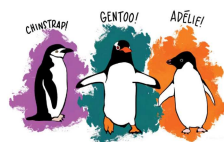
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Our go-to dataset

- In this course, we'll use data to illustrate concepts.
- When possible, we will always begin with the same dataset to minimize cognitive load. - Don't worry; we will explore different datasets as well!
- Let's familiarize ourselves with our main dataset.

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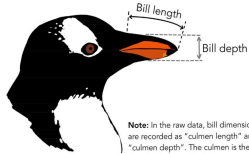
Palmer Penguins Dataset



Artwork by @allison_horst

Dr. Kristen Gorman has collected data on 344 penguins from three islands in the Palmer Archipelago, Antarctica.

Multiple variables were measured about each penguin, such as: **island**, **species**, **bill depth**, **bill length**, **body mass**, **sex**, among others.



Artwork by @allison_horst

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Palmer Penguins Dataset

species	island	bill_length_mm	bill_depth_mm	flipper_length_mm	body_mass_g	sex	year
Adelie	Torgersen	39.1	18.7	181	3750	male	2007
Adelie	Torgersen	39.5	17.4	186	3800	female	2007
Adelie	Torgersen	40.3	18.0	195	3250	female	2007
Adelie	Torgersen	NA	NA	NA	NA	NA	2007
Adelie	Torgersen	36.7	19.3	193	3450	female	2007
Adelie	Torgersen	39.3	20.6	190	3650	male	2007
Adelie	Torgersen	38.9	17.8	181	3625	female	2007
Adelie	Torgersen	39.2	19.6	195	4675	male	2007
Adelie	Torgersen	34.1	18.1	193	3475	NA	2007
Adelie	Torgersen	42.0	20.2	190	4250	NA	2007
Adelie	Torgersen	37.8	17.1	186	3300	NA	2007
Adelie	Torgersen	37.8	17.3	180	3700	NA	2007
Adelie	Torgersen	41.1	17.6	182	3200	female	2007

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Palmer Penguins in R

- The data is available in the package `palmerpenguins`.

- You can install it using
`install.packages('palmerpenguins')`

- Accessing the data:

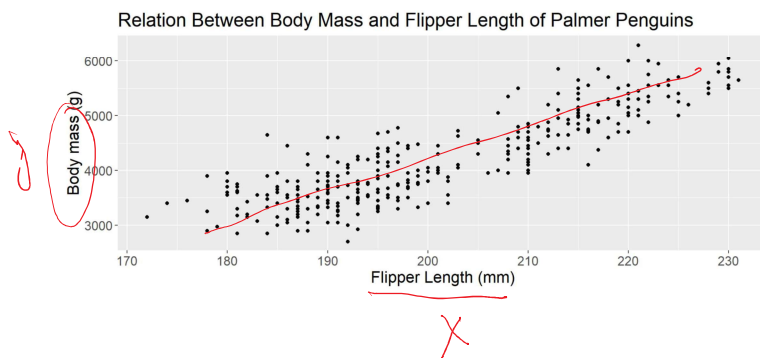
- Load the library `library(palmerpenguins)`.
- The data will be stored in the variables `penguins`.

```
> library(palmerpenguins)
> penguins
# A tibble: 344 × 8
  species island bill_length_mm bill_depth_mm flipper_length_mm
<fct>    <fct>      <dbl>         <dbl>         <int>
1 Adelle  Torgers...  39.1          18.7          181
2 Adelle  Torgers...  39.5          17.4          186
3 Adelle  Torgers...  40.3          18           195
4 Adelle  Torgers...   NA           NA            NA
```

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Flipper Length vs Body Mass

- Suppose we want to use Penguins' **Flipper Length** to explain their **Body Mass**;



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Flipper Length vs Body Mass

- This is a stochastic relation;
 - Many penguins have a flipper length of 190mm , but their weights range from 3050g to 4600g .
 - Other aspects affect body mass besides flipper length.
- The relation is “almost” linear;
- A straight-line model could be a reasonable approximation for this relation.
 - This model is called *Simple Linear Regression*.

maybe
appears to be
likely

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Simple Linear Regression

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Specification

- We have two variables:
 - Response variable (Y): what we are trying to predict/explain;
 - Explanatory Variable (X): the variable used to predict the response;
- We have n observations/samples.
 - Y_i and X_i denote the values of X and Y for observation i .
 - You can think of it as n pairs: $(X_1, Y_1), (X_2, Y_2), \dots, (X_n, Y_n)$.
- For example, for the penguins, i refers to each penguin and can vary from 1 to 344.

i	flipper_length_mm (X)	body_mass_g (Y)
1	181	3750
2	186	3800
3	195	3250
4	NA	NA
...	...	...

- The explanatory variable X is assumed to be fixed for each individual/sample;

data (a sample of size n)

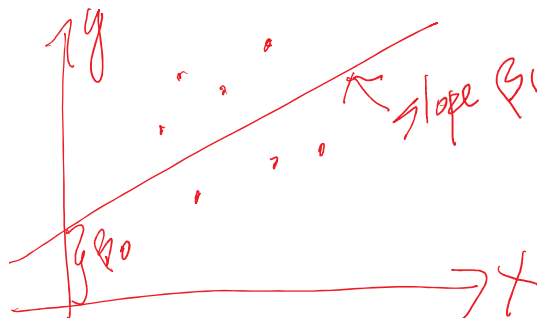
not random

The model

- The model relating X and Y is given by the equation of a line:

$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i, i=1, 2, \dots, n$$

- Let's discuss each of these components in more detail;



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The model: components

- The model relating X and Y is given by the equation of a line:

$$Y_i = \underbrace{\beta_0}_{\text{intercept}} + \underbrace{\beta_1}_{\text{slope}} X_i + \underbrace{\epsilon_i}_{\text{error term}}$$

Handwritten notes:
- A red circle around Y_i with an arrow pointing to it from the word "random".
- A red circle around X_i with an arrow pointing to it from the word "fixed".
- A red circle around ϵ_i with an arrow pointing to it from the word "random".
- A large red arrow points from the ϵ_i term to the text: "the variation in y that cannot be explained by x ".

- Intercept:** tells us the Y value when $X = 0$ (i.e., the value of Y when the line crosses the Y -axis).
- Slope:** tells us how much change in Y to expect for a unit increase in X .
- Error:** captures the variability of the response not explained by the model.

The model: components

- The model relating X and Y is given by:

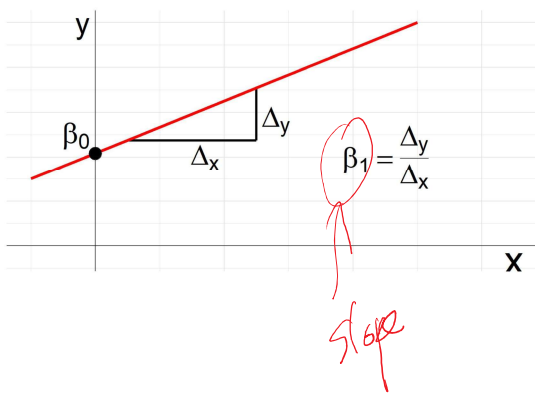
$$Y_i = \underbrace{\beta_0}_{\text{intercept}} + \beta_1 X_i + \epsilon_i$$



The model: components

- The model relating X and Y is given by:

$$Y_i = \underbrace{\beta_0}_{\text{intercept}} + \underbrace{\beta_1}_{\text{slope}} X_i + \epsilon_i$$



Fitting

You might need to refresh this page to show the plot

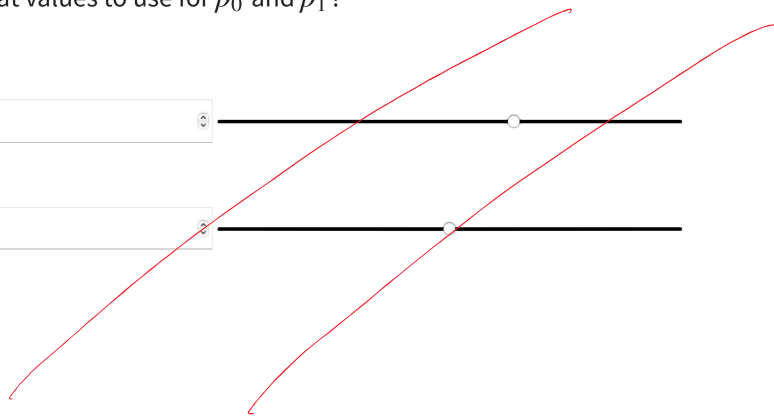
- What values to use for β_0 and β_1 ?

β_0 :

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β_1 :

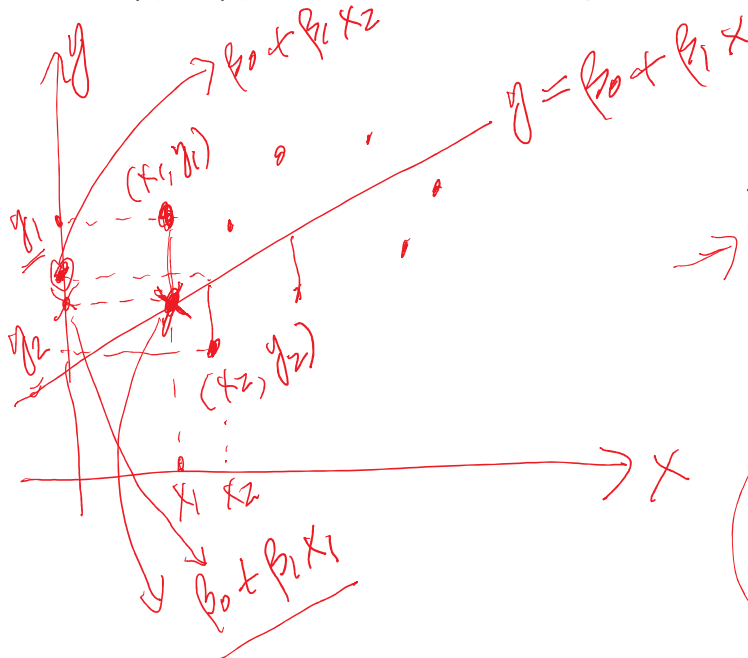
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Fitting

You might need to refresh this page to show the plot

- We want β_0 and β_1 that minimizes the Sum of Square Error;



Q: How to estimate the values of β_0 and β_1 ?

→ You have data first. Then you find a line that is closest to all data points

Q: How to do that?

A: → Minimize $\sum_{i=1}^n (y_i - (\beta_0 + \beta_1 x_i))^2$

The random errors

- The error component, ϵ_i , captures everything that our model

$$\frac{\partial Q}{\partial \beta_0} = 0$$

THE RANDOM ERRORS

- The error component, ε_i , captures everything that our model does not.
- We treat ε_i as a random variable.
 - It has a distribution:
 - We will assume it to be Normal;
 - It has a mean:
 - safely assumed to be 0;
 - It has a variance:
 - unknown and denoted by σ^2 ;
- But what does this mean?

Intuition

True value of β_1 may be between $\hat{\beta}_1 - 1.96 \times SE(\hat{\beta}_1)$ and $\hat{\beta}_1 + 1.96 \times SE(\hat{\beta}_1)$

unbiased estimate
 $E(\hat{\beta}_1) = \beta_1$

Solve $\begin{cases} \frac{\partial Q}{\partial \beta_0} = 0 \\ \frac{\partial Q}{\partial \beta_1} = 0 \end{cases}$

Solutions: $\hat{\beta}_0, \hat{\beta}_1$ are estimates of β_0, β_1 , not the true values of β_0, β_1 (which are never known)

uncertainty of the estimate

Two points: (1) unbiased estimate (2) uncertainty (SE)

The random errors and the response

You might need to refresh this page to show the plot

- Imagine a linear model relating *Height* (m) and *Weight* (kg):

$$\text{Weight} = -166 + 140 \times \text{Height} + \varepsilon$$

• For a 1.63m tall person, we have

$$\text{weight} = -166 + 140 \times \text{Height} + \varepsilon$$

- For a $1.63m$ tall person, we have:

$$\text{Weight} = -166 + 140 \times 1.63 + \varepsilon = 62.2 + \varepsilon$$

estimates

- We would expect this person to weigh around $62.2kg$;
 - But the weight is affected by other factors as well;
 - so we cannot say precisely the value of the weight;
- We have a probability distribution of possible weights for a $1.63m$ tall person:

Height:

1.63

Modelling the average

- For a 1.63m tall person, we have:

$$\text{Weight} = \underline{-166} + 140 \times 1.63 + \varepsilon = 62.2 + \varepsilon$$

Q: Does it still hold for a different dataset?

- We would expect this person to weigh around 62.2kg;
 - Some people will weigh more, and some will weigh less.
- In average, 1.63m tall people weigh 62.2;
- The line $-166 + 140 \times \text{Height}$ gives the mean Weight of people of a given Height;

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The model as conditional expectation

- In general:

$$\underline{E[Y|X]} = \beta_0 + \beta_1 X$$

X is fixed

- This just means that the regression line is the conditional average of Y for a given value of X .

- Note the difference:

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$$

- This equation is for a given point, which is off the line (note the presence of the error term).

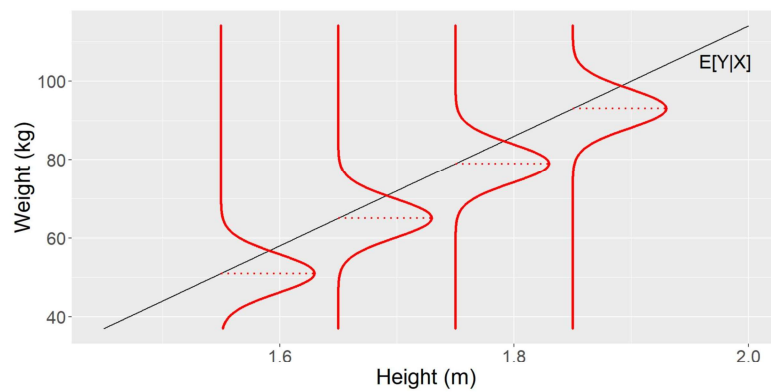
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The model as conditional expectation

$$\begin{array}{c} \text{Point: } Y_i \\ \overbrace{\beta_0 + \beta_1 X_i} \\ \text{Regression Line:} \\ E[Y|X_i] \end{array} + \epsilon_i$$

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The model as conditional expectation



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Interpretation β_0 and β_1

- **Slope:** an increase of 1 unit of X is associated with an expected increase of β_1 units in Y . - It is associated with, not the cause of!
- **Intercept:** The average value of Y when $X = 0$ is β_0 . - Usually, we don't care as much about this parameter.

⚠ Important: Association is not causality.

In general, we **cannot** conclude that changes in X cause a change in Y . The conclusion of causality requires more than a good model.

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Example: Fitting a model

- Let's fit a linear model relating Flipper Length and Body Mass of penguins.
- To fit a linear model in R, we use the `lm` function.
 - Let's explore this function in a code demo!

Code demo - Part I!

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Fitting is estimation

- $E[Y|X] = \beta_0 + \beta_1 X$ is the population's conditional mean for a given value of X .
- But instead of estimating the mean for each value of X separately, which is not feasible, we are assuming a linear structure between the mean of the population and the value of X .
- Therefore, estimating the means for the value of X (in a certain range) reduces to estimate β_0 and β_1 .

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Population vs Sample Regression

- Population Regression:

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$$

- Sample Regression (estimated from the sample):

$$\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_i + \hat{\varepsilon}_i$$

- Note that $\hat{\varepsilon}_i \neq \varepsilon_i$ because $\hat{\beta}_0 \neq \beta_0$ and $\hat{\beta}_1 \neq \beta_1$

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The parameters β_0 and β_1

- Since β_0 and β_1 are parameters, we estimate them based on a sample;
- $\hat{\beta}_0$ and $\hat{\beta}_1$ are the estimators of β_0 and β_1 .
- As statistics, $\hat{\beta}_0$ and $\hat{\beta}_1$ depend on the sample, therefore we will need their **sampling distribution**.

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Inference for $\hat{\beta}_0$

You do not need to memorize these formulae.

- Estimator of β_0 :

$$\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}$$

Inference means what?
Can we generalize the results for a single dataset to a larger population?

Two approaches:

- ① confidence interval for β_0
- ② hypothesis testing for β_0

- Std. Error of β_0 :

$$SE(\hat{\beta}_0) = \sqrt{\sigma^2 \left[\frac{1}{n} + \frac{\bar{X}^2}{\sum_{i=1}^n (X_i - \bar{X})^2} \right]}$$

- This assumes that all errors, ε_i , are uncorrelated with equal the variance σ^2 .

- Sampling distribution of $\hat{\beta}_0$

$$\hat{\beta}_0 \sim N(\beta_0, [SE(\hat{\beta}_0)]^2)$$

- Since σ^2 is unknown, we can estimate using:

$$s^2 = \frac{1}{n-2} \sum_{i=1}^n e_i^2$$

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derived based
on the assumptions

Inference for $\hat{\beta}_1$

You do not need to memorize these formulae.

- Estimator of ρ_1 :

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2}$$

$\hat{\beta}_1 = r \frac{S_y}{S_x}$, where r is the correlation between X and Y . and $S_y^2 = \frac{1}{n-1} \sum (y_i - \bar{y})^2$
 $S_x^2 = \frac{1}{n-1} \sum (x_i - \bar{x})^2$

- Std. Error of $\hat{\beta}_1$:

$$SE(\hat{\beta}_1) = \sqrt{\frac{\sigma^2}{\sum_{i=1}^n (X_i - \bar{X})^2}}$$

- This assumes the variance for all the errors, ϵ_i , is:

- fixed and equal to σ^2
- uncorrelated;

- Sampling distribution of $\hat{\beta}_1$

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Confidence Intervals for β_0 and β_1

You do not need to memorize these formulae.

$$CI(\beta_0, 1 - \alpha) = \hat{\beta}_0 \pm t_{1-\alpha/2}^* \widehat{SE}(\hat{\beta}_0)$$

$$CI(\beta_1, 1 - \alpha) = \hat{\beta}_1 \pm t_{1-\alpha/2}^* \widehat{SE}(\hat{\beta}_1)$$

- Or we can use bootstrap!! Yes, like in STAT 201.

$\rightarrow 1.96$ if $\alpha = 0.05$
 $= z$
 They are derived based on the assumption that ϵ is normally distributed
 Q: What if the normal assumption does not hold?
 A: bootstrap

Example: CI - Penguins data

Code demo - Part 2!

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Example: CI - Penguins data

```
1 # Fit the model
2 penguins_lm <- lm(body_mass_g ~ flipper_length_mm, data = penguins %>% drop
3
4 # Extract the CI
5 confint(penguins_lm, level = 0.95)
```

	2.5 %	97.5 %
(Intercept)	-6482.47	-5261.71
flipper_length_mm	47.12	53.18

- With 95% confidence, we expect an increase in the penguins' weight between 47.12 and 53.18 grams for every 1 mm increase in the penguin's flipper length.

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Example: CI - Penguins data

- Via Bootstrap (slope):

```
1 # Infer package framework
2 ci_slope <-
3   penguins_clean %>%
4     specify(formula = body_mass_g ~ flipper_length_mm) %>%
5     generate(reps = 15000, type = "bootstrap") %>%
6     calculate(stat = "slope") %>%
7     get_ci(type = "percentile", level = 0.95)
```

lower_ci	upper_ci
47.27984	53.10233

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Hypothesis Test

- We can also test the hypothesis that Y and X are linearly related. This is equivalent to testing

$$H_0 : \beta_1 = 0 \quad \text{vs} \quad H_1 : \beta_1 \neq 0$$

- We can change the alternative hypothesis if β_1 cannot be positive or negative.

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Hypothesis Test: Null Model

- Test Statistic and Null model:

$$T = \frac{\hat{\beta}_1 - 0}{\widehat{SE}(\hat{\beta}_1)} \sim t_{n-2}$$

- Reject if p-value $< \alpha$.

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Example: Penguins data

Code demo - Part 3!

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Example: Penguins data

```
1 broom::tidy(penguins_lm, conf.int = TRUE, conf.level = 0.9)
```

term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	-5872.09	310.29	-18.92	0	-6383.90	-5360.29
flipper_length_mm	50.15	1.54	32.56	0	47.61	52.69

- Therefore, we reject the null hypothesis that $\beta_1 = 0$, in favour of $H_1 : \beta_1 \neq 0$.

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Example: range problem

Code demo - Part 4!

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The range problem

- The linear model assumes that the relationship between X and $E[Y|X]$ is linear, which may or may not be true;
- Sometimes, there's a linear association only in part of the data range.
 - The linear model could still be useful when restricted to that specific range;
- We need to exercise caution when using the model outside the range of the data, as the relationship between X and Y may differ significantly.

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Regression vs Correlation analysis

- **Correlation analysis:** we're interested in the strength of linear association between two variables;
 - no distinction between the two variables (no response and no covariate);
 - both variables are assumed to be stochastic;
- **Linear Regression:** we're interested in estimating the conditional average of the response given the value of the covariate. - covariate is assumed to be non-stochastic; - one of the variables is treated as a response and the other as a covariate;

$y = \beta_0 + \beta_1 x + \varepsilon$

$r = \text{Correlation between } X \text{ and } y$

$\hat{\beta}_1 = r \times \frac{s_y}{s_x}$

$s_y > 0$
 $s_x > 0$

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Disclaimer

Warning

All the models in this course are approximations of reality. None of these models will be true or correct, but hopefully they will still be useful.

- There's a famous quote from George E. P. Box:

“Essentially, all models are wrong, but some are useful.”

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Categorical Covariate?

- Note that we can also have a categorical covariate.
- We can try to explain the body mass of Penguins based on the sex of penguins: **male** or **female**.

$$\text{body_mass_g}_i = \beta_0 + \beta_1 \text{sex}_i + \varepsilon_i$$

- But wait! How can we have categories in an equation?

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Dummy variables (2 categories)

- We can encode the categories into multiple variables.
- For example, the variable **sex** could be defined as:

$$\text{sex}_i = \begin{cases} 0 & \text{if penguin } i \text{ is female} \\ 1 & \text{if penguin } i \text{ is male} \end{cases}$$

- A variable **patient_status** could be defined:

$$\text{patient_status}_i = \begin{cases} 0 & \text{if patient } i \text{ is healthy} \\ 1 & \text{if patient } i \text{ is sick} \end{cases}$$

- If we have a variable with two categories, we need only 1 dummy variable to represent it.

Back to the model

- We can try to explain the body mass of Penguins based on the sex of penguins: **male** or **female**.

$$\text{body_mass_g}_i = \beta_0 + \beta_1 \text{sex}_i + \varepsilon_i$$



Where is the line?

- Note that there is no line in this case.
- Sex** cannot be 0.1 or 0.5. It can only be 0 or 1.
- So, what is going on?

$$\text{body_mass_g}_i = \begin{cases} \beta_0 + \beta_1 + \varepsilon_i & \text{if penguin } i \text{ is male} \\ \beta_0 + \varepsilon_i & \text{if penguin } i \text{ is female} \end{cases}$$

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We are just comparing means

- Remember that in regression, we model the mean given the value of a covariate.
- So, in this case, we are modelling the mean of female penguins and the mean of male penguins;
- Note that:
 - β_0 is the average `body_mass_g` of female penguins.
 - β_1 is the **difference** in means.
 - $\beta_0 + \beta_1$ is the average `body_mass_g` of male penguins.
- To test if the means are equal, we need to test $H_0 : \beta_1 = 0$ vs $H_1 : \beta_1 \neq 0$.

Testing for equality

```
1 penguin_sex_lm <- lm(body_mass_g ~ sex, data = penguins_clean)
2 tidy(penguin_sex_lm)
```

term	estimate	std.error	statistic	p.value
(Intercept)	3862.27	56.83	67.96	0
sexmale	683.41	80.01	8.54	0

- The term **sexmale** is β_1 in our model.
 - The **female** sex is the reference level, and its average is given by the intercept: 3862.27.
 - The estimated difference of **body_mass_g** between male and female is 683.41.
 - The estimated average for **male** $3862.27 + 683.41 = 4545.68$

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Equivalent to t-test

```
1 t_test_average <- t.test(body_mass_g ~ sex,
2                           var.equal = TRUE, data = penguins_clean)
3
4 tidy(t_test_average) %>%
5   select(estimate, estimate1, estimate2, statistic, p.value)
```

estimate	estimate1	estimate2	statistic	p.value
-683.41	3862.27	4545.68	-8.54	0

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R for us

- R is pretty good dealing with categorical variables.

- All we need to do is to use factors, e.g.,

```
1 penguins_clean %>%  
2   mutate(sex = as_factor(sex))
```

- The `lm` will create the dummy variables and tell us the levels of the factor associated with the coefficient.